正十七边形的尺规作图

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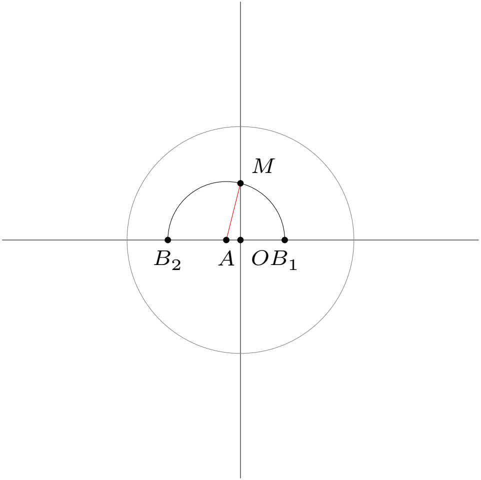
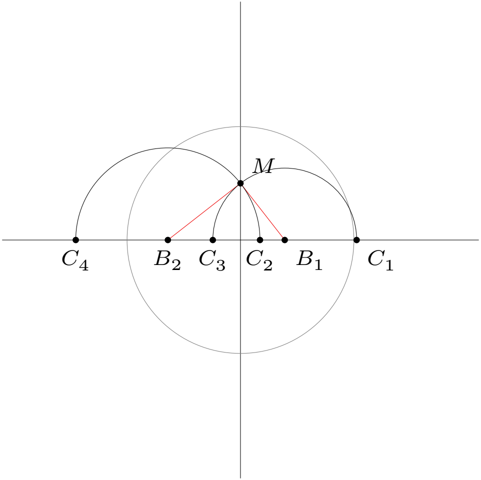
- 作单位圆 O
- $A\left(-\frac{1}{8},0\right)$,$M\left(0,\frac{1}{2}\right)$
- 以 A 为圆心，AM 为半径作圆弧，交 x 轴于 B_1, B_2

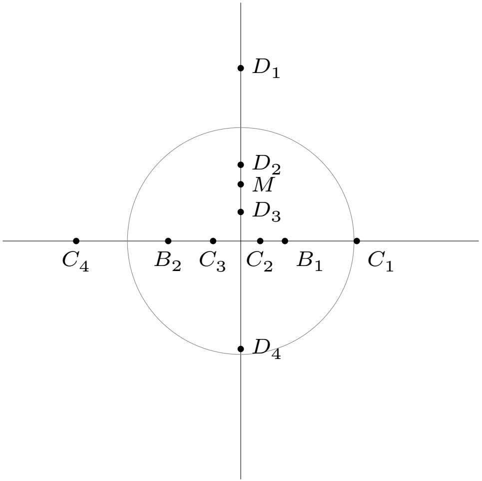


分别以 B_1,B_2 为圆心，B_1M 和 B_2M 为半径作圆弧，交 x 轴于 C_1,C_2,C_3,C_4

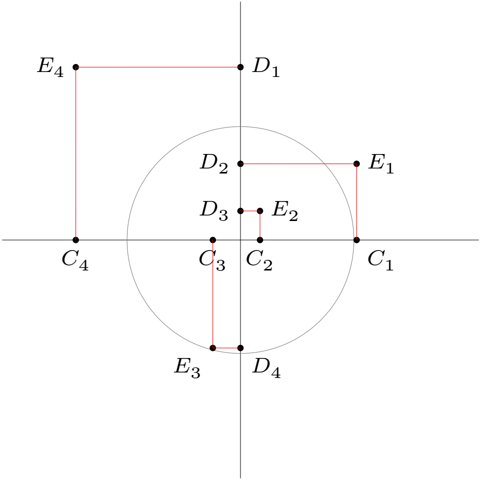


在 y 轴上，作出 D_1,D_2,D_3,D_4，使得

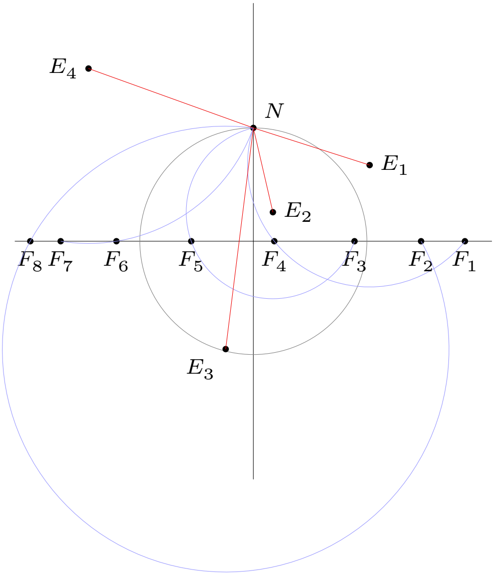
$$MD_1 = OC_1$$
$$MD_2 = OC_2$$
$$MD_3 = OC_3$$
$$MD_4 = OC_4$$



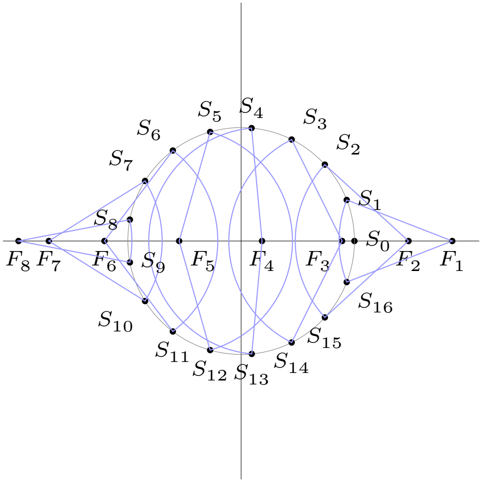
作出如图所示矩形，标记顶点E_1,E_2,E_3,E_4.



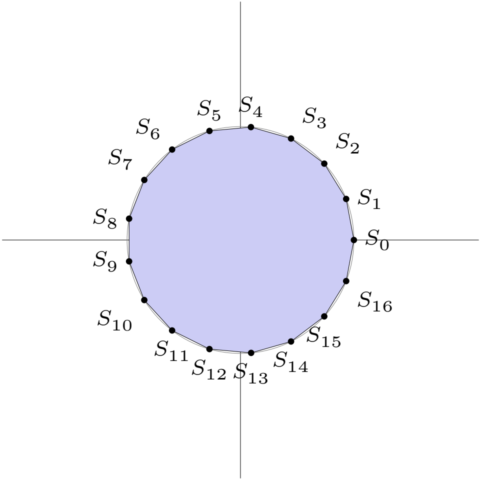
- $N(0,1)$
- 分别以 $E_{1\sim4}$ 为圆心，E_1N,E_2N,E_3N,E_4N 为半径作圆弧，交 x 轴于 $F_{1\sim8}$.



- $S_0(1,0)$
- 分别以 $F_{1\sim8}$ 为圆心，作单位圆（弧），交圆 O 于 16 个点 $S_{1\sim16}$



依次连接 17 个点，便作出了正十七边形.



参考文献

https://www.zhihu.com/question/26096850/answer/2333073721